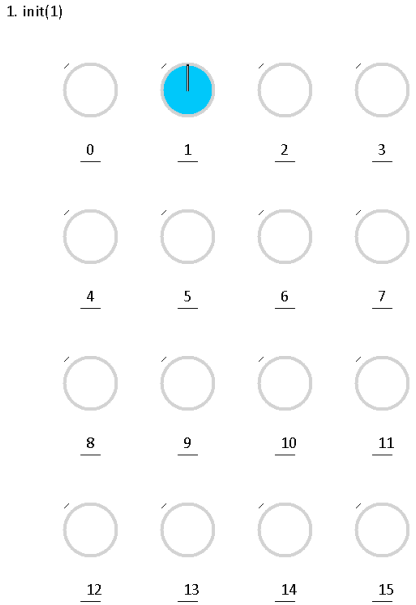
1. init(1)

```
QBit
 #  v
 0: 1 >_

 1: 0 >_

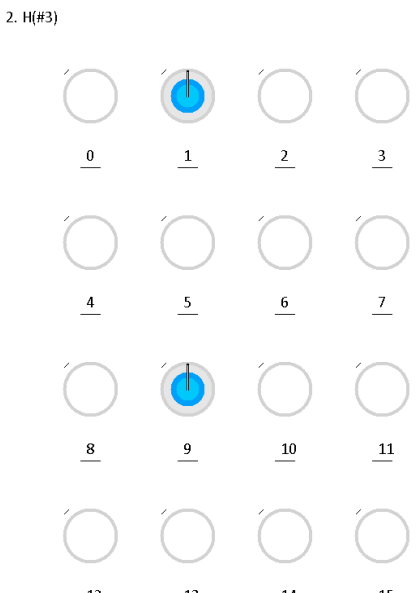
 2: 0 >_

 3: 0 >_
```



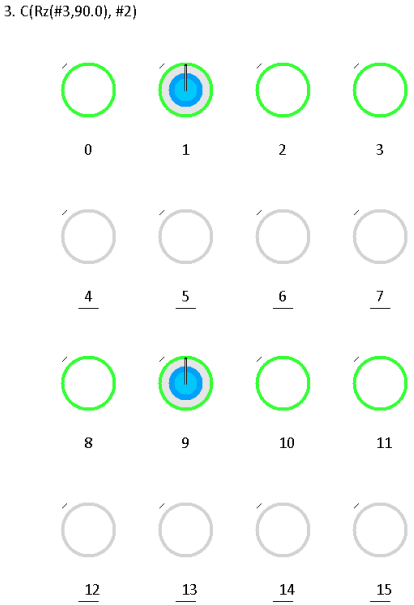
2. H(#3)

```
QBit
 #  v
 0: 1 >_|_____________
        |
 1: 0 >_|_____________
        |
 2: 0 >_|_____________
        |
 3: 0 >_|__H__________
        |
       L QFT : qft
```



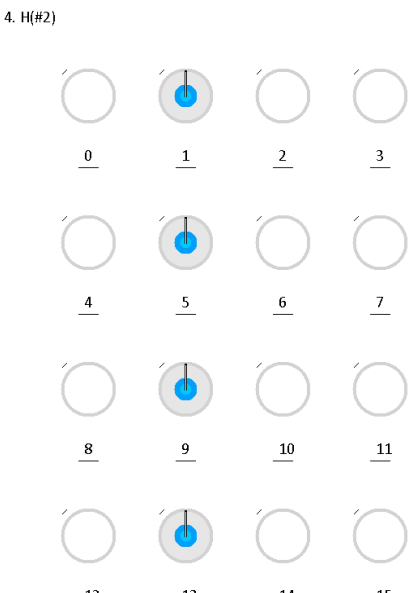
3. C(Rz(#3,90.0), #2)

```
QBit
 #  v
 0: 1 >_|_____________
        |
 1: 0 >_|_____________
        |
 2: 0 >_|_____._______
        |     |
 3: 0 >_|__H_Rz_____
        |      90
       L QFT : qft
```



4. H(#2)

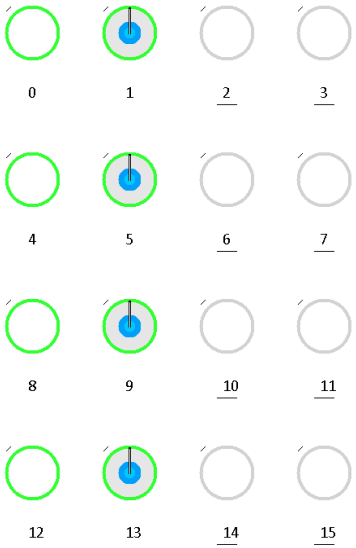
```
QBit
 #  v
 0: 1 >_|_____________
        |
 1: 0 >_|_____________
        |
 2: 0 >_|_____._____H_
        |     |
 3: 0 >_|__H_Rz_______
        |      90
       L QFT : qft
```



5. C(Rz(#3,45.0), #1)

```
QBit
 #  v
 0:..._|_________________
       |
 1:..._|_____________._
       |               |
 2:..._|____._____H_|_
       |    |        |
 3:..._|__H_Rz______Rz
       |      90      4
      L QFT : qft
```

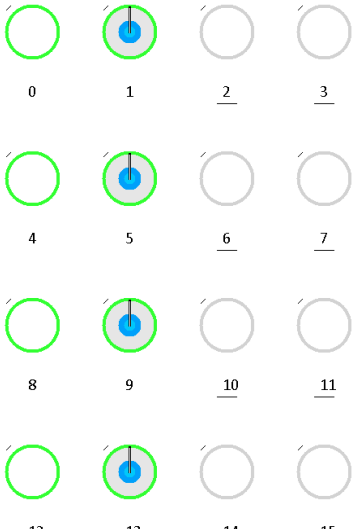
5. C(Rz(#3,45.0), #1)



6. C(Rz(#2,90.0), #1)

```
QBit
 #  v
 0:..._____________________
 1:..._________._____._
               |     |
 2:...._____H_|_____Rz
      |       |      9
 3:...Rz______Rz______
       90      45
       : qft
```

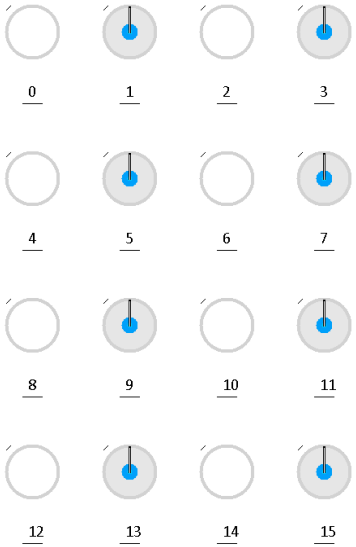
6. C(Rz(#2,90.0), #1)



7. H(#1)

```
QBit
 #  v
 0:...____________________
 1:...______.______.___
           |      |
 2:...____H_|_____Rz__
           |       90
 3:...______Rz_________
      0      45
       qft
```

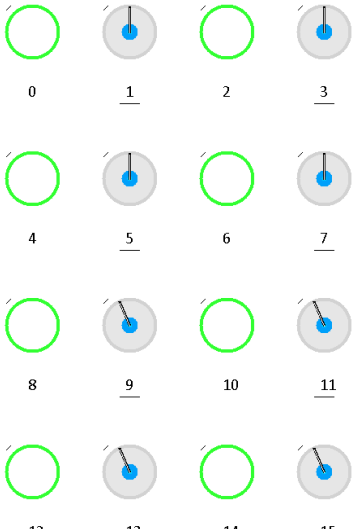
7. H(#1)



8. C(Rz(#3,22.5), #0)

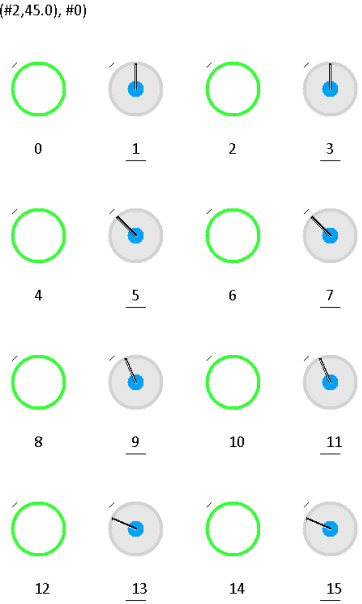
```
QBit
 #  v
 0:...____________________._
                           |
 1:....______._____H_|_
      |      |             |
 2:...|_____Rz_____|_
      |       90    |
 3:...Rz____________Rz
       45            2
```

8. C(Rz(#3,22.5), #0)



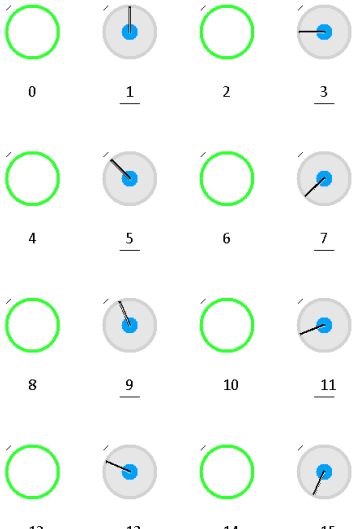
9. C(Rz(#2,45.0), #0)

```
QBit
 #  v
 0:...________._____._
               |     |
 1:....____H_|_____|_
     |         |     |
 2:...Rz_______|____Rz
       90      |      4
 3:...________Rz_____
              22.5
```

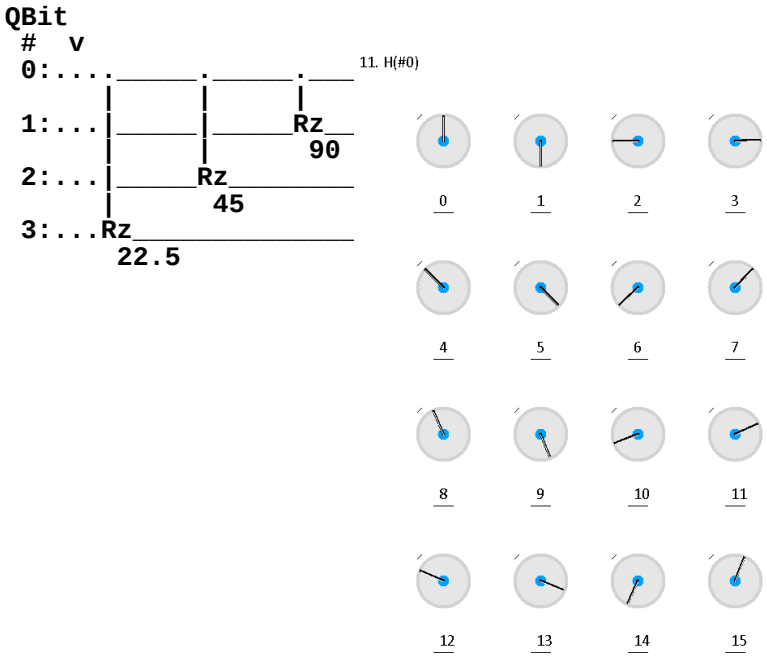


10. C(Rz(#1,90.0), #0)

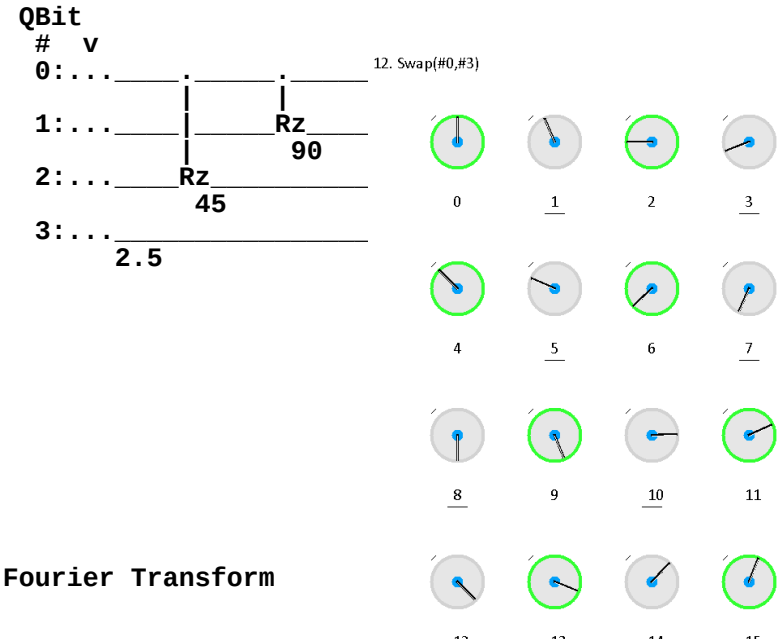
```
QBit
 #  v
 0:...__._____._____._
       |     |     |
 1:...H_|_____|____Rz
       |     |      9
 2:...__|____Rz_____
       |      45
 3:...__Rz___________
        22.5
```



11. H(#0)

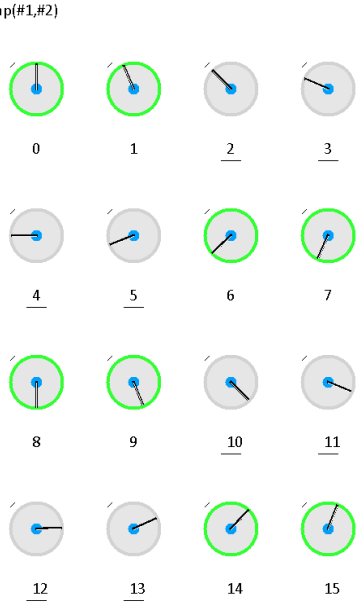


12. Swap(#0,#3)



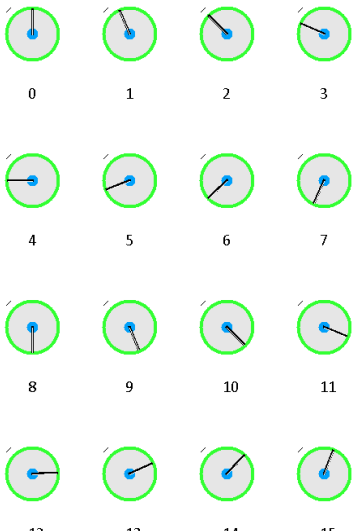
13. Swap(#1,#2)

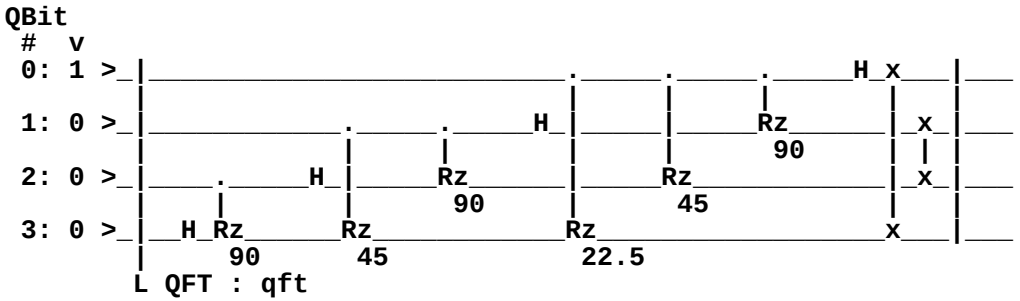
```
QBit
 #  v
 0:...__._____._____H_
        |       |
 1:...__|_____Rz_____
        |      90
 2:...__Rz____________
       45
 3:...________________
      5
```



14. F(QFT qft)

```
QBit
 #  v
 0:...____._____H_x___
        |       |
 1:...____Rz_____|_x_
         90     |  |
 2:...__________|_x_
      5         |
 3:...__________x___
```




QBit
v
0: 1 >
1: 0 >
2: 0 >
3: 0 >
H
Rz
90
45
22.5
x
L QFT : qft